



NNP Pocket Guide 2

FEN, CV, and Resp

Version 3, October 2026

Companion to the procedures booklet; same ring.

Sections

-  FEN: GIR, TPN, calories, bilirubin
-  CV: pressors, profiles, BP norms
-  Resp: blood gas, vent
-  Ref: sources

Starting points, not orders. Follow unit protocol and your attending.



GIR (mg/kg/min)

Shortest	$\%D \times \text{mL/kg/day} \div 144$
Hourly	$\%D \times \text{mL/hr} \div (6 \times \text{kg})$ $= \%D \times \text{mL/hr} \times 0.167 \div \text{kg}$
Long way	$\text{mL/day} \times \%D \text{ (decimal)} = \text{g}; \times 1000 = \text{mg}; \div \text{kg} \div 1440$

%D for a target GIR = $\text{GIR} \times 144 \div \text{mL/kg/day}$

Example: D10 at 80 mL/kg/day = GIR 5.6.

Targets

Start 4–6, then titrate to glucose; preterm infants need at least 4.

Max dextrose

12.5% peripheral. 25% central with the tip in the SVC.



TPN: start and advance

When: <32 wk, VLBW/ELBW, GI anomaly or surgery, NPO >3–5 days.

Day 1: D10 at 60–80 mL/kg/d, AA 2–3 g/kg/d, lipid 1 g/kg/d.

Amino acids	Within 24 h at 2–3; +0.5/d to 3–4, max 4
Lipid	1–2 → 3 g/kg/d. 0.5–1 prevents EFAD. TG >200: stop or cut 0.5–1.
Sodium	None first 24 h, then 2–4 mEq/kg/d
Potassium	None first 24–48 h, then 1–3 mEq/kg/d once urine output is established
Ca gluconate	200–400 mg/kg/d

Peripheral TPN under 900 mOsm/L.



TPN calories

Glucose	3.4 kcal/g
Protein	4 kcal/g
Fat	9 kcal/g
20% lipid	1 g = 5 mL = 10 kcal (2 kcal/mL)

$$\text{Total} = \text{GIR} \times 5 + \text{AA g/kg} \times 4 + \text{lipid g/kg} \times 10$$

Exact dextrose kcal/kg/day = %D $\times$ mL/kg/day $\times$ 0.034 (GIR $\times$ 4.9).

$$\text{Lipid g/kg/day} = \text{mL/hr} \times 24 \times 0.2 \div \text{kg}$$

Example: 1.8 kg, D12 at 10.2 mL/hr, AA 3.5, lipid 1.1 mL/hr $\approx$ 99 kcal/kg/day.

Goals

Preterm 110–130, term 100–120 kcal/kg/day. Gain 15–20 g/kg/day.

$$\text{Feeds: kcal/mL} = \text{kcal/oz} \div 30$$

20 $\rightarrow$ 0.67	22 $\rightarrow$ 0.73	24 $\rightarrow$ 0.8
26 $\rightarrow$ 0.87	28 $\rightarrow$ 0.93	30 $\rightarrow$ 1.0



Bilirubin

35 weeks and up

Thresholds are hour-specific curves by GA and risk: plot them in BiliTool or PediTools.

Neurotoxicity risk factors: GA <38 wk, albumin <3.0, hemolysis (isoimmune, G6PD), sepsis, instability in the past 24 h.

Escalation: TSB within 2 of the exchange line = emergency. NICU, intensive phototherapy, PO + IV fluids, STAT labs, TSB every 2 h.

Stop phototherapy at ≥ 2 below the start threshold; rebound in ~24 h (6–12 h if started before 48 h).

Under 35 weeks (Maisels 2012)

PMA	Photo	Exchange
<28 wk	5–6	11–14
28–29	6–8	12–14
30–31	8–10	13–16
32–33	10–12	15–18
34–35	12–14	17–19

TSB mg/dL. Born <35 wk: stay on the top row after 35 wk PMA.



Pick the pressor

BP = CO × SVR: pipes, pump, or both?

Low diastolic	Pipes (low SVR) → vasopressor ($\alpha 1$ or V1)
Low systolic	Pump (low contractility) → inotrope ($\beta 1$ or PDE3)
Both	Cold, mottled, acidotic → both actions

	Warm (low SVR)	Cold (high SVR)
Skin	Warm, pink	Cool, mottled
Refill	Flash <2 s	>3 s
Pulses	Bounding	Thready
Pulse pressure	Wide	Narrow
Causes	Sepsis, NEC, big PDA	Cold sepsis, transitional, post-ligation

Hypotensive, warm, and bounding is **not** reassuring: it's vasodilated and needs tone.



Pressor doses

Drug	Dose	Use
Dopamine	≤ 2 / ≥ 5 –10 / ≥ 10 mcg/kg/min	DA $\rightarrow$ β_1 $\rightarrow$ α_1 with dose. ≥ 10 : R $\rightarrow$ L ductal shunt risk. Not with PPHN.
Dobutamine	≥ 3 –15 mcg/kg/min	β_1 inotrope, mild β_2 dilation. Avoid in outflow obstruction (IDM).
Epinephrine	0.02–0.1, then ≥ 0.1 (caution >0.2)	Low dose β , high dose α_1 . Lactate, glucose, tachycardia.
Norepinephrine	0.02–0.4 mcg/kg/min	$\alpha_1 > \beta_1$. Severe vasoplegia.
Vasopressin	0.0002–0.005 units/kg/min	V1a. Catecholamine-resistant; spares PVR. Low Na, low platelets.
Milrinone	0.25–0.75 mcg/kg/min, no load	PDE3 inodilator. Low output + PH, post-ligation. Onset 1–2 h; drops BP.
Hydrocortisone	1 mg/kg IV q8h	Adrenal insufficiency, or refractory on ≥ 2 agents.



Decision tree and algorithm

1. **Bed loose** (warm, low diastolic) → norepinephrine (or dopamine)
2. **Pump weak** (cool, poor contractility) → dobutamine
3. **Pump weak + high afterload** (PPHN, post-op) → milrinone, add a pressor if BP sags
4. **Both failing** → epinephrine
5. **Still low on catecholamines** → vasopressin; think hydrocortisone

Around it

Perfusion adequate (NIRS, urine output, lactate, echo)? Then permissive hypotension. Reassess after every change; BP up with worse perfusion means over-constriction. Wean catecholamines first, keep milrinone while PH persists, wean vasopressin last.



Hemodynamic profiles

Read diastolic BP, pulse pressure, perfusion index, SVR, cardiac output, and StO₂ together.

Profile	Pattern
Vasodilatory	↓ DBP, ↓ SVR; ↑ PP, PI, CO (warm shock)
Vasoconstrictor	↑ DBP, ↑ SVR; ↓ PP, PI, CO (cold shock)
Cardiogenic	↓ CO, low PP, PI, DBP
L-R shunt	↑ CO, ↑ PP; ↓ DBP, ↓ SVR (PDA run-off)
Volume depletion	↓ CO, ↓ PP; ↑ SVR

Same low diastolic, opposite fixes: vasodilatory needs a vasoconstrictor; volume-depleted needs volume.

Epinephrine 0.2 mcg/kg/min for an hour $\approx$ one resuscitation bolus (10 mcg/kg).



BP norms (after 2 weeks)

By PMA; 50th / 95th / 99th percentile, mmHg (Dionne 2012, corrected table).

PMA	MAP	Systolic	Diastolic
44	63/80/85	88/105/110	50/68/73
42	62/76/81	85/98/102	50/65/70
40	60/75/80	80/95/100	50/65/70
38	59/74/79	77/92/97	50/65/70
36	57/72/77	72/87/92	50/65/70
34	50/65/70	70/85/90	40/55/60
32	49/64/69	68/83/88	40/55/60
30	48/63/68	65/80/85	40/55/60
28	45/58/63	60/75/80	38/50/54
26	38/57/63	55/72/77	30/50/56

High BP: $\geq$ 95th on 3 occasions; $>$ 99th, investigate. **"MAP $\geq$ GA":** weak evidence; judge with perfusion. **Cuff:** compare MAP (the measured value), right arm.



Read a blood gas

1. **pH:** <7.35 acidemic, >7.45 alkalemic
2. **PaCO_2 :** respiratory?
3. **HCO_3^- / **BE:** metabolic?**
4. **Compensation, then the bedside picture**

CO_2 opposite pH $\rightarrow$ respiratory. HCO_3^- with pH $\rightarrow$ metabolic.

pH	7.35–7.45 (>7.30 may be OK in context)
PaCO_2	35–45
HCO_3^-	22–26
BE	-2 to +2; worse than -4 to -6, watch

Uncompensated	pH off, other value normal
Partial	pH off, both abnormal
Compensated	pH normal, both abnormal

Lungs fix CO_2 quickly; kidneys fix HCO_3^- over days. High CO_2 **and** low HCO_3^- = mixed acidosis, not compensation.



What moves what

CO₂ ↑	pH ↓ (respiratory acidosis)
CO₂ ↓	pH ↑ (respiratory alkalosis)
HCO₃⁻ ↓ (lactate, TPN acid, RTA, diarrhea)	pH ↓, BE more negative
HCO₃⁻ ↑ (vomiting, diuretics, bicarb, acetate)	pH ↑
Bicarb given	HCO ₃ ⁻ + H ⁺ → carbonic acid → CO₂ + water : CO ₂ rises; pH can fall if not ventilated
CO₂ high for days	Kidneys keep HCO ₃ ⁻ → pH toward normal
Metabolic acidosis	Lungs blow off CO ₂ quickly
Rate or tidal volume ↑	CO ₂ ↓, pH ↑
MAP or FiO₂ ↑	Oxygenation ↑

Bicarb: only with adequate ventilation; not in the same line as catecholamines; can lower K.



Vent modes

AC	Every breath supported. Poor effort, apnea, extreme prematurity.
SIMV	Synced breaths plus spontaneous. Weaning, low support, sedation backup.
Volume guarantee	Hits a set volume; unstable compliance; guards against volutrauma.
Pressure control	Caps pressure; volume varies. RDS, severe BPD.
HFOV	Open lung, tiny volumes. PIE, air leak, PPHN.
HFJV	Fast small bursts; conventional vent gives PEEP. PIE, refractory CO ₂ .

By disease

RDS: CPAP → PC or volume-targeted; HFOV if severe. **TTN:** HFNC or CPAP. **BPD:** long expiratory time, permissive CO₂, often VG. **PIE/air leak:** HFJV or HFOV. **Pulmonary hemorrhage:** higher MAP. **PPHN:** conventional or HFOV.



What to change: conventional and HFOV

Conventional

CO₂ high	Rate +5–10; PIP +2 for chest rise; lower PEEP if over-expanded
O₂ low	FiO ₂ ↑; PEEP ↑; PIP +1–2; Ti 0.4–0.6 s
Weaning	FiO ₂ to 80% before changing PIP, I:E, PEEP; steps PIP 2, FiO ₂ 5%, rate 5

$$\text{MAP} = (\text{PIP} - \text{PEEP}) \times \text{Ti} \div (\text{Ti} + \text{Te}) + \text{PEEP}$$

HFOV

CO₂ high	Amplitude ↑; lower Hz if needed
CO₂ low	Amplitude ↓; raise Hz
O₂ low	MAP ↑; FiO ₂ ↑
O₂ high	FiO ₂ ↓, then MAP ↓

Lower Hz = bigger volumes, more CO₂ removal; Hz changes are usually a senior call.



Jet, weaning, and OI

HFJV

Ventilation: PIP above PEEP (ΔP). Oxygenation: PEEP (+1–2 when $FiO_2 \geq 0.6$). Don't wean PEEP until $FiO_2 < 0.35$ –0.40.

HFOV weaning

- $PaCO_2 < 50$ and $pH > 7.25$: wean amplitude
- $FiO_2 < 70\%$, $PaO_2 > 60$, good inflation: wean MAP
- Extubation: term MAP 11–14, amp ≤ 20 ; LBW MAP 10–12, ≤ 15 ; VLBW MAP < 10 , < 12
- Air leak: wean MAP hard, smallest amplitude, accept CO_2 high 50s–low 60s, PIE side down

$$OI = MAP \times FiO_2 \times 100 \div PaO_2$$

OI > 40 for > 2 h, or in 3 of 5 gases: associated with about 80% mortality.



Sources

Course lectures and handouts: Intro to Neonatal FEN; Neonatal TPN; FEN formulas; Chorioamnionitis and EOS screening; Cardiovascular Function and Treatments; Choosing a Pressor; Neonatal Hemodynamics; Module 2 Respiratory; Blood Gas Interpretation. Program NNP Brain reference; NNP course quick-reference card.

Bilirubin: AAP 2022 Clinical Practice Guideline (Kemper et al.); Maisels et al., J Perinatol 2012; UNC NCCC hyperbilirubinemia guideline (2025).

BP: Dionne, Abitbol, Flynn 2012 (erratum) via Batton, Clin Perinatol 2020; Starship and Medscape neonatal hypertension.

Vent: Loyola neonatal ventilation supplement; SPA oscillator review; UNC NCCC high-frequency ventilation; UW Madison NICU jet guideline; NHS GGC HFOV.

Bicarb: Canterbury DHB and NSW ANMF neonatal guidelines.

Full versions, calculators, and links are in the NNP app.